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Cryptographic Primitives

2.1 Symmetric Algorithms

2.1.1 Stream Cipher Algorithms

Accepted stream cipher algorithms:

- SNOW 2.0 (ISO/IEC 18033-4)
 - . 128-bit key length for MODERATE.
 - . 256-bit key length for ADVANCED.
- SOSEMANUK¹ (from eSTREAM project)
 - . 128-bit and 256-bit key lengths for MODERATE.
 - . Not accepted for ADVANCED.

Common notes:

- Initialization Vector (IV) must be at least 128 bits.
- Stream ciphers must be used with a different IV for each key.
- A key must be used only once.
- Valid decryption must never be relied on for authenticity.

2.1.2 Block Ciphers Algorithms

Accepted block cipher algorithms:

- Advanced Encryption Standards (AES) as in FIPS-197
 - . 128-bit and 192-bit key lengths for MODERATE.
 - . 256-bit key length for ADVANCED.
- Camellia (ISO/IEC 18033-3)
 - . 128-bit and 192-bit key lengths for MODERATE.
 - . 256-bit key length for ADVANCED.
- Serpent²
 - . 128-bit and 192-bit key lengths for MODERATE.
 - . 256-bit key length for ADVANCED.

¹ C. Berbain *et al.* "Sosemanuk, a Fast Software-Oriented Stream Cipher." In: Robshaw M., Billet O. (eds.) New Stream Cipher Designs. LNCS 4986. Springer, 2008.

² E. Biham, R. Anderson, and L. Knudsen. SERPENT: A new block cipher proposal. In Fast Software Encryption - FSE'98, volume 1372 of Lecture Notes in Computer Science, pages 222–238. Springer-Verlag, 1998.

2.2 Asymmetric Algorithms

Accepted asymmetric algorithms:

- RSA
 - . At least 3072 bits for key length and “ e ” value > 65537 for MODERATE.
 - . Not accepted for ADVANCED.
 - . Strong primes must be used.
- Diffie-Hellman and Finite Field
 - . At least 3072 bits for key length and “ q ” value (subgroup size) of 256 for MODERATE.
 - . Not accepted for ADVANCED.
- Elliptic Curve Discrete Logarithm Problem based Algorithms (ECDLP)
 - . NIST P-256, NIST P-384, BrainpoolP256r1, BrainpoolP384r1 and Curve25519 for MODERATE.
 - . NIST P-521, Curve448³ and BrainpoolP512r1 for ADVANCED.

2.3 Hash Functions

Accepted hash functions:

- Secure Hash Algorithm 2 (SHA-2)
 - . SHA2-384 and SHA2-512/256 for MODERATE.
 - . Not accepted for ADVANCED.
- Secure Hash Algorithm 3 (SHA-3)
 - . SHA3-256, SHA3-384, SHAKE128 and SHAKE256 for MODERATE.
 - . SHA3-512 for ADVANCED.

Common notes:

- Hash functions must be inversion-resistant, pre-image resistant and collision resistant.
- For SHAKE128, the output size “ d ” value must be ≥ 256 bits.
- For SHAKE256, the output size “ d ” value must be ≥ 512 bits.
- SHA2-384 and SHA2-512/256 are sometimes referred to as SHA-384 and SHA-512/256 respectively.

2.4 Lightweight Crypto Algorithms

Accepted lightweight algorithms (on limited systems with constrained resources, where the use of conventional cryptographic standards is not applicable):

- Block ciphers (ISO/IEC 29192-2)
 - . PRESENT, 80-bit or 128-bit key lengths.

³ Curve448 is accepted for ADVANCED even though it operates at 224-bit security level because of its increased performance, resistance to a wide-range of side channel attacks and ease of implementation.

- . CLEFIA, 128-bit, 192-bit or 256-bit key lengths.
- Stream ciphers (ISO/IEC 29192-3)
 - . Enocoro, 80-bit or 128-bit key lengths.
 - . Trivium, 80-bit key length.
- Asymmetric mechanisms (ISO/IEC 29192-4)
 - . Unilateral authentication mechanisms.
 - . ALIKE key exchange.
 - . Identity-based signature.
- Hash functions (ISO/IEC 29192-5)
 - . PHOTON, 80-bit, 128-bit, 160-bit, 224-bit or 256-bit output sizes.
 - . SPONGENT, 88-bit, 128-bit, 160-bit, 224-bit or 256-bit output sizes.
 - . Lesamnta-LW, 256-bit output size.
- Message Authentication Code (MAC) as in ISO/IEC 29192-6
 - . Tsudik's keymode, hash-based.
 - . Chaskey12, 128-bit key length.